

SQL ANALYSIS REPORT

Adventure works Data Analysis

Adventure works

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Adventure Works Project: Business Questions

Q1. Overview

Q2. Which product categories generate the highest revenue and profit?

Q3. Which product subcategories have high sales but low profit margins?

Q4. How has monthly revenue changed over time and where are the major growth or decline periods?

Q5. Which territories contribute the most revenue and profit?

Q6. Who are the top 10 customers by lifetime value (LTV)?

Q7. What percentage of total revenue comes from the top 20% of customers?

Q8. Which customer segments generate the highest average order value?

Q9. Which products generate the highest profit, not just revenue?

Q10. Which products has low sales in the top category

Q11. Which territories show the highest revenue growth rate year-over-year?

Q12. Are there territories with high order volume but low profitability?

Q1 OVERVIEW OF THE DATA

```
1 use adventureworks_project;
2 select
3 count(*) as total_orders,
4 round(sum(subTotal),2) as total_sales,
5 round(avg(subTotal),2) as average_order_value
6 from SalesOrderHeader;
7
8 select count(*) as total_products from Product;
9 select count(*) as total_customers from Customer;
10
11 select
12 sum(((sod.UnitPrice*(1 - sod.UnitPriceDiscount))
13 - p.StandardCost)*sod.OrderQty)
14 as total_profit
15 from
16 SalesOrderDetail as sod
17 left join product as p
18 on sod.ProductID = p.ProductID;
19
```

	total_orders	total_sales	average_order_value
1	31465	109846381.40	3491.07
	total_products		
1	504		
	total_customers		
1	19820		
	total_profit		
1	9371903.7694		

DATASET SUMMARY :

- Total Orders: 31,465
- Total Sales: 109.8M
- Average Order Value: 3,491
- Total Products: 504
- Total Customers: 19,820
- Total Profit: 9.37M

Q2 Which product categories generate the highest revenue and profit?

```
1 use AdventureWorks_Project;
2 select
3 pc.Name,
4 sum(sod.UnitPrice * (1 - sod.UnitPriceDiscount)*
5 sod.OrderQty) as total_sales,
6 sum((sod.UnitPrice * (1 - sod.UnitPriceDiscount) - p.StandardCost) *
7 sod.OrderQty) as total_profit,
8 sum((sod.UnitPrice * (1 - sod.UnitPriceDiscount) - p.StandardCost) *
9 sod.OrderQty)
10 /sum(sod.UnitPrice * (1 - sod.UnitPriceDiscount) * sod.OrderQty)*100 as margin
11 from SalesOrderDetail as sod
12 inner join Product as p on sod.ProductID = p.ProductID
13 inner join ProductSubcategory as ps on p.ProductSubcategoryID = ps.ProductSubcategoryID
14 inner join ProductCategory as pc on ps.ProductCategoryID = pc.ProductCategoryID
15 group by pc.Name
16 order by total_sales desc;
```

	Name	total_sales	total_profit	margin
1	Bikes	94651172.7442	7936394.0554	8.38
2	Components	11802593.2737	490233.1704	4.15
3	Clothing	2120542.6258	308882.2681	14.56
4	Accessories	1272072.8992	636394.2755	50.02

Observation :

1. Bikes had the highest sales (94.6M).
2. Bikes generated the highest profit (7.9M).
3. Accessories had the best profit margin (50%).
4. Clothing performed moderately in both sales and profit.

Q3 Which product subcategories have high sales but low profit margins?

```
2 use AdventureWorks_Project;
3
4 select
5 ps.Name,
6 (sum(sod.LineTotal)) as sales,
7 sum((sod.UnitPrice * (1 - sod.UnitPriceDiscount) - p.StandardCost) * sod.OrderQty) as profit,
8 (sum(sod.LineTotal))/ sum((sod.UnitPrice * (1 - sod.UnitPriceDiscount)
9 - p.StandardCost) * sod.OrderQty) as profit_margin
10 from SalesOrderDetail as sod
11 left join Product as p
12 on sod.ProductID = p.ProductID
13 left join ProductSubcategory as ps
14 on p.ProductSubcategoryID = ps.ProductSubcategoryID
15 group by ps.Name
16 -- having sum((sod.UnitPrice * (1 - sod.UnitPriceDiscount) - p.StandardCost) * sod.OrderQty) < 0
17 order by sales desc;
```

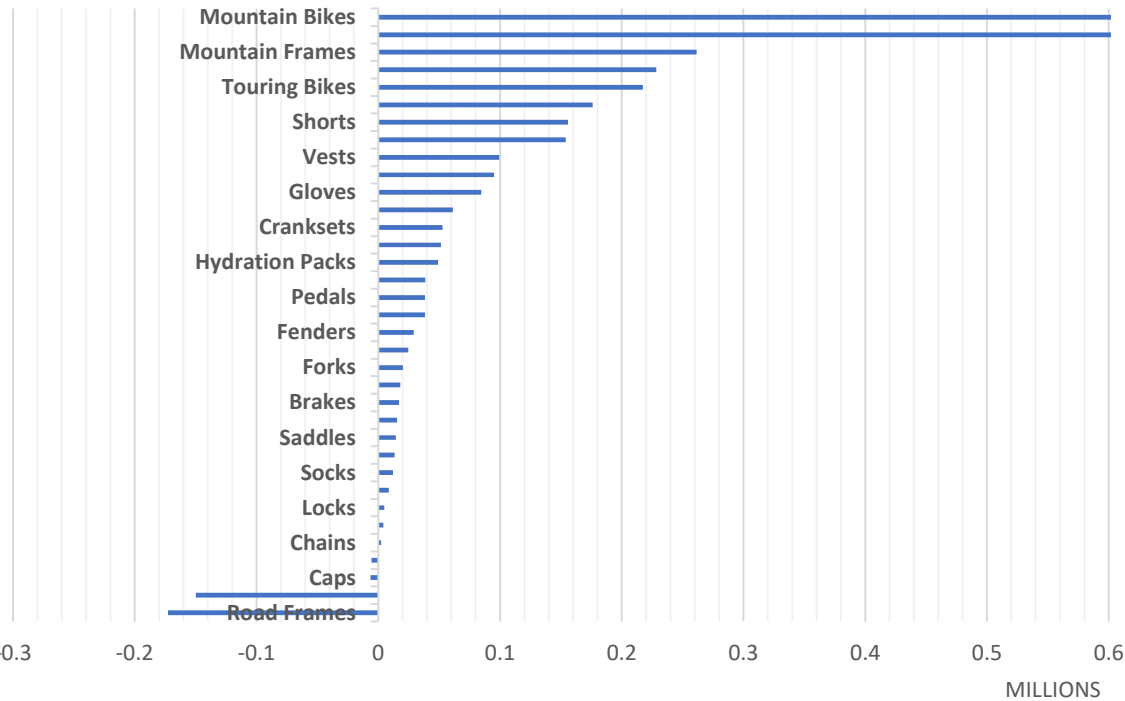
	Name	sales	profit	profit_margin
1	Road Bikes	43909437.508212	2811074.4171	15.620161
2	Mountain Bikes	36445443.937380	4908041.8978	7.425658
3	Touring Bikes	14296291.259139	217277.7405	65.797311
4	Mountain Frames	4713930.229240	261604.8037	18.019280
5	Road Frames	3851350.600747	-172617.2908	-22.311499
6	Touring Frames	1642327.686185	-5573.4108	-294.671924
7	Jerseys	752259.388034	-149887.1505	-5.018838
8	Wheels	680831.354061	176044.8657	3.867374
9	Helmets	484048.529277	228329.1424	2.119959
10	Shorts	413600.513342	155974.105	2.651725

Sub categories with negative profit

	Name	profit
1	Jerseys	-149887.1505
2	Caps	-6301.7922
3	Touring Frames	-5573.4108
4	Road Frames	-172617.2908

Q3 Which product subcategories have high sales but low profit margins?

Profit by Category



KEY FINDINGS :

On The Analysis I observed That Road Bikes Have Highest Sales But Low Profit Margin (15%) And I Also observed There is Good Profit Margin From Touring Bikes(65%)

We should Focus On :

Few products with Are No profit They Are In Negative

1. Jerseys (-149,887)
2. Caps (-6,301)
3. Touring Frames (-5573)
4. Road Frames (-172,617)

Q4 How has monthly revenue changed over time and where are the major growth or decline periods?

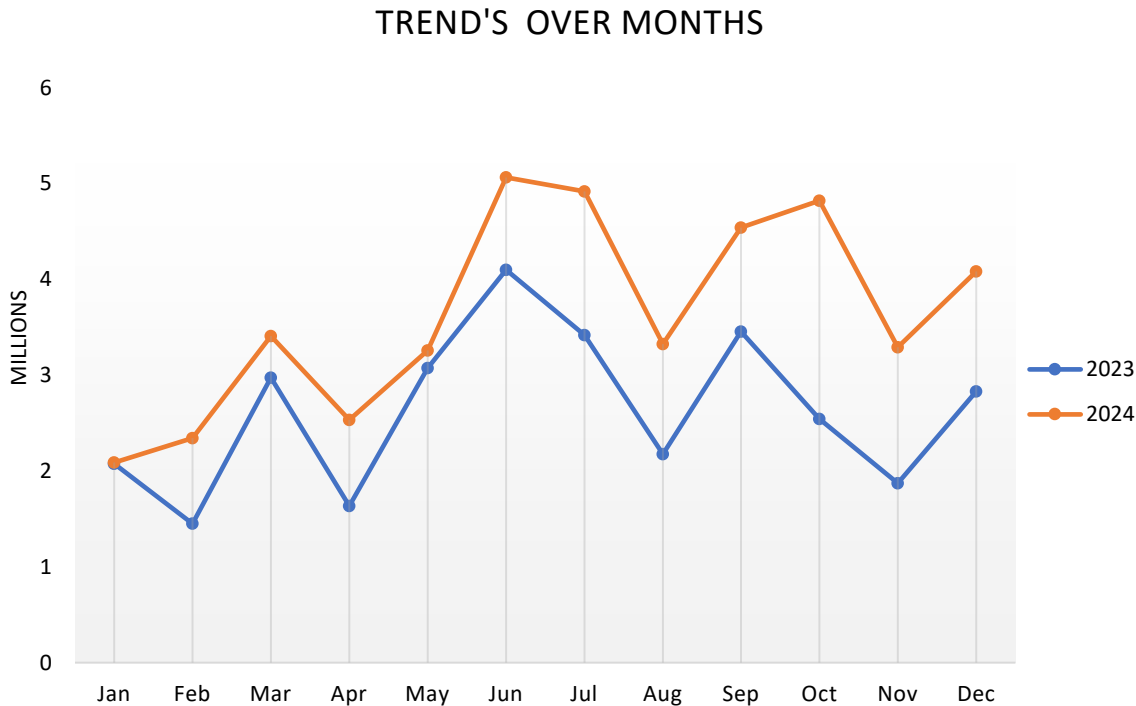
```
1 use AdventureWorks_Project;  
2  
3 select  
4 year(orderDate) as year_,  
5 month(orderDate) as month_,  
6 sum(subTotal) as total_sales  
7 from SalesOrderHeader  
8 group by year(orderDate),month(orderDate)  
9 order by year_ asc, month_ asc;
```

	year_	month_	total_sales
5	2022	9	2836409.6133
6	2022	10	2269438.2267
7	2022	11	1437342.5339
8	2022	12	2504196.209
9	2023	1	2076791.6086
10	2023	2	1449883.2018
11	2023	3	2975748.2385
12	2023	4	1634600.7984
13	2023	5	3074602.8138
14	2023	6	4099354.3572
15	2023	7	3417953.8699
16	2023	8	2175637.218
17	2023	9	3454151.9405
18	2023	10	2544091.1056
19	2023	11	1872701.9762

Note: This analysis on next Slide uses only 2023 and 2024 data. The years 2022 and 2025 were excluded because they do not contain data for all months.

Contd..

Q4 How has monthly revenue changed over time and where are the major growth or decline periods?



Observation :

- 1 Sales was low in **February** and **April**.
- 2 Sales increased steadily from **May** and reached a peak in **June**.
- 3 Strong sales performance was maintained during **June** and **July**.
- 4 Sales declined in **August**
- 5 Sales recovered and performed well in **September** and **October**.
- 6 Another drop was observed in **November**
- 7 Sales improved again in **December**
- 8 Overall, the highest sales were recorded in **June, July, September, and October**.

Q5 Which territories contribute the most revenue and profit?

```
1  -- Which territories contribute the most revenue and profit
2  use AdventureWorks_Project;
3  SELECT
4  st.name,
5  SUM(sod.LineTotal) as revenue ,
6  sum(sod.Profit) as profit
7
8  FROM SalesOrderHeader as soh
9  JOIN SalesOrderDetail as sod
10 ON soh.SalesOrderID = sod.SalesOrderID
11 JOIN SalesTerritory as st
12 ON soh.TerritoryID = st.TerritoryID
13 group by st.name
14 order by revenue desc,profit desc;
```

Results		Messages	
	name	revenue	profit
1	Southwest	24184609.600810	1563696.32
2	Canada	16355770.454862	404304.97
3	Northwest	16084942.547585	1364591.56
4	Australia	10655335.959317	3434244.04
5	Central	7909009.005872	-142451.33
6	Southeast	7879655.072151	-151057.77
7	United Kingdom	7670721.035475	1231901.59
8	France	7251555.646926	893088.75
9	Northeast	6939374.481005	-269493.90
10	Germany	4915407.595885	1043011.06

Results		Messages	
	name	revenue	profit
1	Central	7909009.005872	-142451.33
2	Southeast	7879655.072151	-151057.77
3	Northeast	6939374.481005	-269493.90

KEY INSIGHTS :

1. Most revenue and profit generated by southwest with Revenue **(\$24.18M)** and the highest profit **(\$1.56M)**
2. On The Second Place We have Canada with Revenue **(\$16.35M)** and the profit **(\$0.40M)**

We should focus on :

1. There are Three Territories With negative Profit But This Territories have Good Revenue

Q6 Who are the top 10 customers by lifetime value (LTV)?

```
1  |-- Who are the top 10 customers by lifetime value (LTV)?
2  use AdventureWorks_Project;
3  select
4  Top 10
5  c.CustomerID,
6  sum(soh.SubTotal) as total_sales
7  from SalesOrderHeader as soh
8  left join Customer as c
9  on soh.CustomerID = c.CustomerID
10 group by c.CustomerID
11 order by total_sales desc;
```

Results		Messages
	CustomerID	total_sales
1	29818	877107.1923
2	29715	853849.1795
3	29722	841908.7707
4	30117	816755.5763
5	29614	799277.895
6	29639	787773.0438
7	29701	746317.5293
8	29617	740985.8338
9	29994	730798.7139
10	29646	727272.6493

Key Insights :

Customer ID 29818 is the top customer, generating \$877K in total sales.

The top 3 customers each contributed more than \$840K in sales.

All top 10 customers generated \$727K in total sales.

Q7 What percentage of total revenue comes from the top 20% of customers?

```
1  -- What percentage of total revenue comes from the top 20% of customers?
2  Use AdventureWorks_Project;
3  DECLARE @top20 INT = (SELECT COUNT(*) * 0.2 FROM Customer);
4
5  select
6  ((select sum(total_sales)
7  from
8      (select Top (@top20)
9      c.CustomerID,|
10      sum(soh.SubTotal) as total_sales
11      from SalesOrderHeader as soh
12      left join Customer as c
13      on soh.CustomerID = c.CustomerID
14      group by c.CustomerID
15      order by total_sales desc)t) /sum(soh.subTotal)) * 100
16  as total_revenue_generated_by_top_20_percent_people
17  from SalesOrderHeader as soh;
```

Results		Messages
total_revenue_generated_by_top_20_percent_people		
1	90.05	

Key Insight :

On the Analysis I observed That 90 % Revenue Was generating by the Top 20 % people

Q8 Which customer segments generate the highest average order value

```
1  -- Which customer segments generate the highest average order value
2  use AdventureWorks_Project;
3  select
4  c.CustomerSegment,
5  AVG(soh.SubTotal) as average_order_value
6  from SalesOrderHeader as soh
7  left join Customer as c
8  on soh.CustomerID = c.CustomerID
9  group by c.CustomerSegment ;
```

Results		Messages
	CustomerSegment	average_order_value
1	Individual	3491.0656

Conclusion: Individual customers had an average order value of 3491.07. Store customers had no orders, so they were not included in the analysis.

Q9 Which products generate the highest profit, not just revenue?

```
1  -- Which products generate the highest profit, not just revenue?
2  use AdventureWorks_Project
3
4  select |
5  p.Name,
6  sum(sod.Profit) as total_profit ,
7  sum(sod.LineTotal) as revenue
8  from SalesOrderDetail as sod
9  left join product as p
10 on sod.ProductID = p.ProductID
11 group by p.Name
12 order by total_profit desc , revenue desc;
```

	Name	total_profit	revenue
1	Mountain-200 Black, 42	674218.38	4009494.761841
2	Mountain-200 Black, 38	673446.33	4400592.800400
3	Mountain-200 Black, 46	666742.34	3309673.216908
4	Mountain-200 Silver, 38	663784.32	3693678.025272
5	Mountain-200 Silver, 46	629643.45	3434256.941928
6	Mountain-200 Silver, 42	611084.22	3438478.860423
7	Road-150 Red, 48	470356.53	1540803.062000
8	Road-150 Red, 62	466321.81	1769096.688000
9	Road-150 Red, 52	421112.27	1415563.612000
10	Road-150 Red, 56	406080.86	1847818.628000
11	Road-150 Red, 44	391565.66	1340419.942000

Key Insights

Mountain-200 Black,42 generated the highest total profit of 674,218.38, making it the most profitable product
On the Second Place We have Mountain-200 Black,38 With Total profit of 673,446.33

Q10 Which products has low sales in the top category

```
1  -- Which products has low sales in the top category
2  use AdventureWorks_Project;
3  select
4  p.Name , sum(sod.LineTotal) as sales_by_product
5  from SalesOrderDetail as sod
6  left join Product as p
7  on sod.ProductID = p.ProductID
8  left join ProductSubcategory as ps
9  on p.ProductSubcategoryID = ps.ProductSubcategoryID
10 where ProductCategoryID = 1
11 group by p.Name
12 order by sum(sod.LineTotal) asc
```

Results		Messages
	Name	sales_by_product
1	Mountain-500 Black, 52	96982.204000
2	Mountain-500 Black, 40	101734.116000
3	Mountain-500 Silver, 44	122512.431600
4	Mountain-500 Black, 44	125925.668000
5	Touring-3000 Yellow, 58	130898.575500
6	Touring-3000 Blue, 44	133365.033375
7	Touring-3000 Blue, 62	135284.008125
8	Mountain-500 Black, 42	136293.476000
9	Road-450 Red, 48	136467.864000
10	Mountain-500 Silver, 42	141360.498000
11	Mountain-500 Silver, 48	142897.270800
12	Mountain-500 Silver, 40	145089.432000
13	Mountain-500 Black, 48	157569.082000

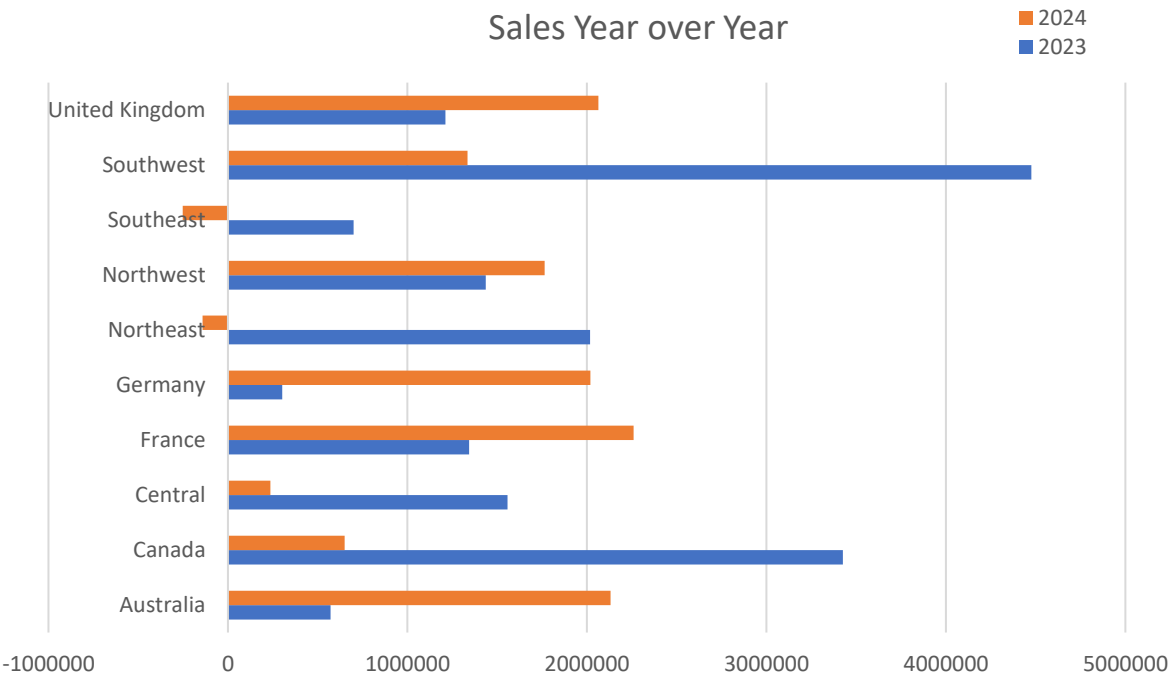
Conclusion: The Mountain-500 Black, 52 generated the lowest sales among all bike products. These low-selling products may need better marketing, discounts to Improve Sales

Q11 Which territories show the highest revenue growth rate year-over-year?

```
1  -- Which territories show the highest revenue growth rate year-over-year?
2  use AdventureWorks_Project;
3  select
4  st.name,
5  year(soh.OrderDate) as year_,
6  sum(soh.SubTotal) as sales,
7  LAG(sum(soh.SubTotal)) over(partition by st.name
8  order by year(soh.orderDate) ) as prev_sales,|
9  -----
10 ( (sum(soh.SubTotal) - LAG(sum(soh.SubTotal))
11 over(partition by st.name
12 order by year(soh.orderDate) ) )) as difference_
13 from
14 SalesOrderHeader as soh
15 left join SalesTerritory as st
16 on soh.TerritoryID = st.TerritoryID
17 where year(soh.OrderDate) < 2025
18 group by st.Name , year(soh.OrderDate)
19 ORDER BY st.Name, YEAR(soh.OrderDate)
```

Q11 Which territories show the highest revenue growth rate year-over-year?

	name	year_	sales	prev_sales	difference_
1	Australia	2022	1542891.1152	NULL	NULL
2	Australia	2023	2114048.3731	1542891.1152	571157.2579
3	Australia	2024	4246450.5549	2114048.3731	2132402.1818
4	Canada	2022	2154591.8432	NULL	NULL
5	Canada	2023	5580848.0456	2154591.8432	3426256.2024
6	Canada	2024	6231209.9635	5580848.0456	650361.9179
7	Central	2022	1200662.2317	NULL	NULL
8	Central	2023	2758256.6569	1200662.2317	1557594.4252
9	Central	2024	2994225.3834	2758256.6569	235968.7265
10	France	2022	213817.7584	NULL	NULL
11	France	2023	1557152.937	213817.7584	1343335.1786
12	France	2024	3816543.3278	1557152.937	2259390.3908
13	Germany	2022	246860.5448	NULL	NULL
14	Germany	2023	550070.7367	246860.5448	303210.1919
15	Germany	2024	2570269.3486	550070.7367	2020198.6119
16	Northeast	2022	756233.2602	NULL	NULL
17	Northeast	2023	2773503.8298	756233.2602	2017270.5696
18	Northeast	2024	2631259.5991	2773503.8298	-142244.2307
19	Northwest	2022	2817381.9112	NULL	NULL
20	Northwest	2023	4254635.1897	2817381.9112	1437253.2785



Observation:

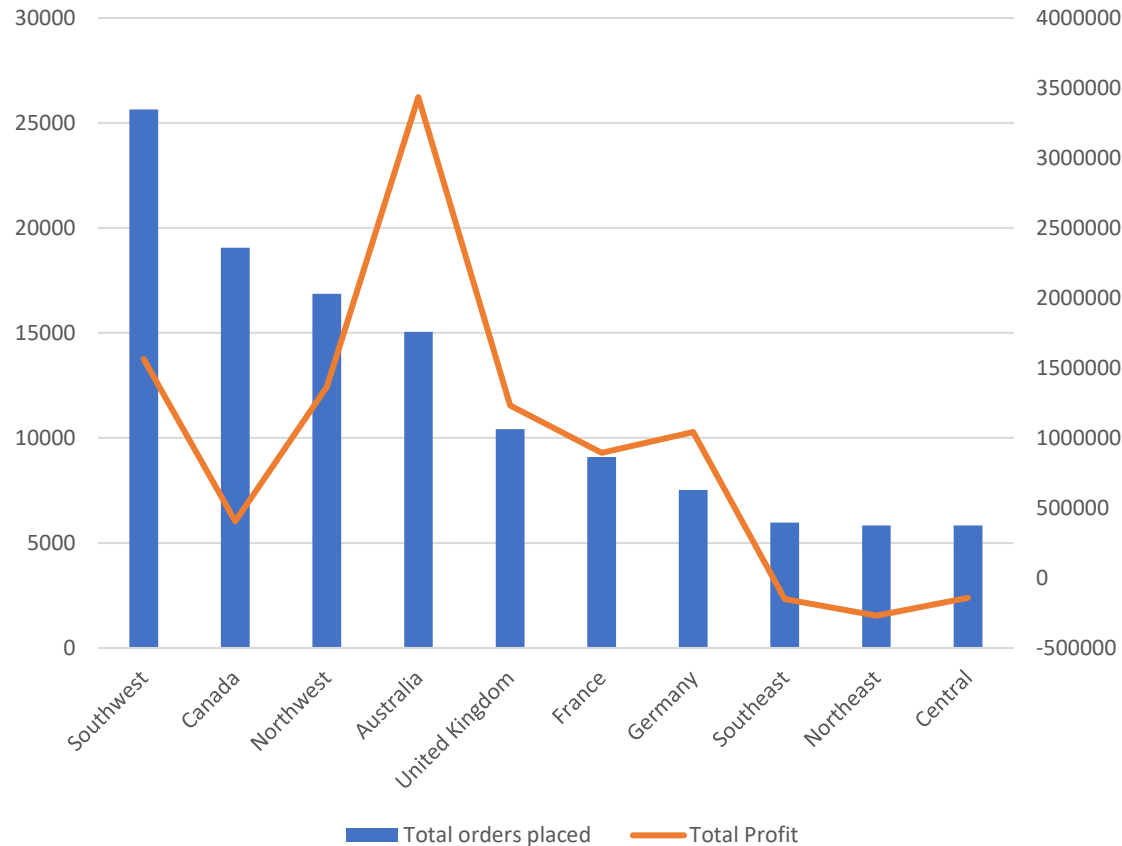
Most territories showed good revenue growth year over year. Australia, Germany, the United Kingdom, Northwest, and France had good growth. However, Canada and Central showed a decrease in sales compared to the previous year. Northeast and Southeast also had negative growth, which means their sales went down. These territories need more attention to understand why sales decreased and how they can be improved.

Q12 Are there territories with high order volume but low profitability?

```
1  -- Are there territories with high order volume but low profitability?
2  use AdventureWorks_Project;
3  select
4  st.name,
5  count( distinct soh.SalesOrderDetailID) as total_orders_placed,
6  sum(soh.Profit) as total_profit|
7  from SalesOrderDetail as soh
8  left join  SalesOrderHeader as sod
9  on soh.SalesOrderID =sod.SalesOrderID
10 left join SalesTerritory as st
11 on sod.TerritoryID =st.TerritoryID
12 group by st.Name
13 order by total_orders_placed desc;
```

	name	total_orders_placed	total_profit
1	Southwest	25644	1563696.32
2	Canada	19064	404304.97
3	Northwest	16865	1364591.56
4	Australia	15058	3434244.04
5	United Kingdom	10426	1231901.59
6	France	9088	893088.75
7	Germany	7528	1043011.06
8	Southeast	5976	-151057.77
9	Northeast	5836	-269493.90
10	Central	5832	-142451.33

Q12 Are there territories with high order volume but low profitability?



Observation:

Southwest, Canada, and Northwest had the highest number of orders, but their profit was lower compared to Australia. Australia had fewer orders but generated the highest profit. The United Kingdom and France also showed good profit with fewer orders.

We should focus on :

Southeast, Northeast, and Central has negative profit. These territories We should focus



Analysis Complete

This SQL documentation covers 12 Business Questions

12

Queries Analyzed